

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Upstream Gas Gathering Concept Select & Revalidation Report

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Revision History

Rev No.	Date	Description of Revision
R00	31 Jul 2022	Issued for Interdisciplinary Checks
R01	01 Aug 2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

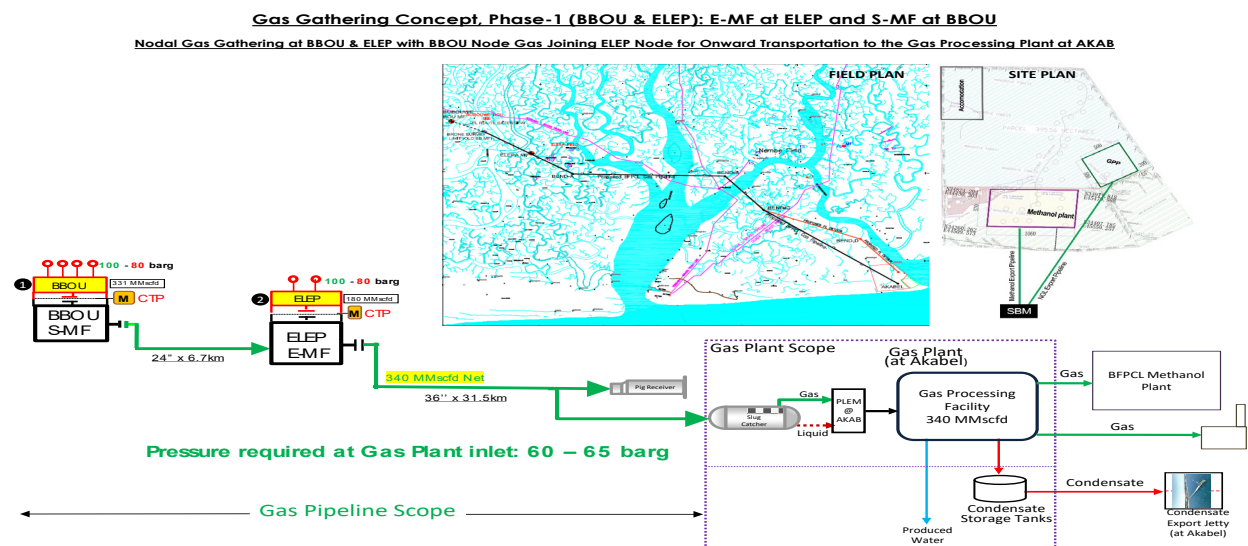


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to re-validate the proposed gas gathering concept and pipeline sizes established during the concept select studies for the total nodes production of the 1 Bscfd, and the scenarios for phased development and gas intake.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Strategic Objective

The strategic objective will follow the basic principle of optimal mix of line sizes and compression requirement to deliver the gas at the required pressure at destination, and in this case the following steps:

1. A configuration of pipeline system that will deliver the gas to the Gas Plant at the required pressure of 60 – 65 barg across the GSPA band of flowing tubing head pressure of 100 – 80 barg, without the need for booster compressor. This is usually the preferred option owing to both CAPEX and life cycle cost in the case of a mix with compression.
2. A rational mix of pipeline size and compression, albeit the higher capex and life cycle cost. The deployment of Compressor is a last resort reactive to reservoir pressure depletion recovery when it is economical to do so, which could happen under the uncertainties of the production forecast; or with an overwhelming increase in throughput that will reduce the arrival pressure as will be the case with 1 BSCFD case in situations where the theoretical pipeline size becomes prohibitive for implementation, as is the case for 1 BSCFD case.

Hence:

3. Sizing of the Phase-1 pipelines to deliver the gas at the required pressure, without the need for booster compressor, with a practical margin that will take care of uncertainties and errors in parameters. As these Phase-1 pipelines will also be applicable to phase-2, their sizing is cross-checked with the phase-2 requirements.
4. Re-verification that the Phase-2 pipelines Sizing will fit the phase-2 production, as well as robust enough to take in and deliver the 1 BSCFD to the Evacuation Manifold at ELEP, from where it will be delivered to AKAB Gas Hub through the ELEP – AKAB main line and the loop line. Delivering the 1 BSCFD to AKAB at the same pressure as the Phase-2 may not be feasible. The objective at this point is therefore to deliver the gas at a reasonable pressure that will make compression economic. The phase 1 & 2 pipelines are already sized to their minimum diameters to avoid compression and so are already contributing positively to this aspired reasonable pressure.
5. The combination of Pipe Flow Hydraulics Analysis of the Pipeline Network, and Flow Assurance was key to this re-validation, and the exercise was done referencing the CSR / FDP production forecast as basis for the construction of instantaneous forecasts used for the process simulations.

4. Production Forecast

The Nodal Forecasts used for the analysis is shown below. The forecast was constructed from the uprated DCQ 270, in the absence of the FDP production forecast from SPDC. The forecast has now been received and as benchmark the constructed nodal forecast is rather more optimistic which is quite ok for the analysis.

4.1. The Nodal Production Forecast

BBOU TO ELEPA PIPELINE SEGMENT (24" x 6.7km)									
year	BBOU NODE			BUBOUWE BOU			OPUGBENE		
	TOTAL FLUID RATE								
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
0	181	944	907	181	944	907	0	0	0
1	181	878	907	181	878	907	0	0	0
2	181	788	907	181	788	907	0	0	0
3	181	742	7,855	181	742	7,855	0	0	0
4	181	707	7,547	181	707	7,547	0	0	0
5	215	840	8,491	215	840	8,491	0	0	0
6	305	1,217	9,002	305	1,217	9,002	0	0	0
7	327	1,323	8,193	327	1,323	8,193	0	0	0
8	331	1,338	8,038	331	1,338	8,038	0	0	0
9	93	1,763	2,527	67	239	2,397	26	1,525	129
10	75	3,242	1,301	21	59	1,031	54	3,183	270
11	74	3,457	1,074	16	42	784	58	3,416	290
12	71	4,084	1,513	2	4	85	69	4,080	1,429
13	71	4,161	3,048	0	0	0	71	4,161	3,048
14	94	5,541	3,748	0	0	0	94	5,541	3,748
15	154	9,082	3,859	0	0	0	154	9,082	3,859
16	170	12,115	3,148	0	0	0	170	12,115	3,148
17	174	16,226	1,667	0	0	0	174	16,226	1,667
18	177	15,812	1,558	0	0	0	177	15,812	1,558
19	182	15,606	2,775	0	0	0	182	15,606	2,775

ELEPA TO AKABEL PIPELINE SEGMENT (36" x 31.5km)															
year	ELEPA NODE			BUBOUWE BOU			ELEPA			NUN RIVER			SEIB		
	TOTAL FLUID RATE														
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	361	11,032	1,807	181	944	907	180	10,088	900	0	0	0	0	0	0
2	361	10,965	1,807	181	878	907	180	10,088	900	0	0	0	0	0	0
3	361	10,876	1,807	181	788	907	180	10,088	900	0	0	0	0	0	0
4	361	10,830	15,835	181	742	7,855	180	10,088	7,980	0	0	0	0	0	0
5	361	10,794	15,213	181	707	7,547	180	10,088	7,666	0	0	0	0	0	0
6	360	8,985	13,393	215	840	8,491	145	8,146	4,902	0	0	0	0	0	0
7	357	4,128	11,599	305	1,217	9,002	52	2,911	2,597	0	0	0	0	0	0
8	356	2,953	9,647	327	1,323	8,193	29	1,630	1,454	0	0	0	0	0	0
9	356	2,724	9,275	331	1,338	8,038	25	1,386	1,236	0	0	0	0	0	0
10	382	18,828	4,837	67	239	2,397	19	1,076	960	244	14,464	1,221	26	1,525	129
11	385	21,637	3,357	21	59	1,031	11	628	561	245	14,583	1,225	54	3,183	270
12	385	21,933	3,015	16	42	784	9	478	426	245	14,583	1,225	58	3,416	290
13	386	22,799	13,832	2	4	85	1	52	46	245	14,583	10,844	69	4,080	1,429
14	386	22,905	16,513	0	0	0	0	0	0	245	14,583	10,417	71	4,161	3,048
15	384	21,996	14,149	0	0	0	0	0	0	196	10,914	6,653	94	5,541	3,748
16	379	22,323	11,246	0	0	0	0	0	0	71	4,159	3,528	154	9,082	3,859
17	380	26,558	8,272	0	0	0	0	0	0	40	2,329	1,976	170	12,115	3,148
18	382	34,433	5,014	0	0	0	0	0	0	34	1,980	1,680	174	16,226	1,667
19	380	33,161	4,421	0	0	0	0	0	0	26	1,538	1,304	177	15,812	1,558
20	379	32,111	6,313	0	0	0	0	0	0	15	898	762	182	15,606	2,775

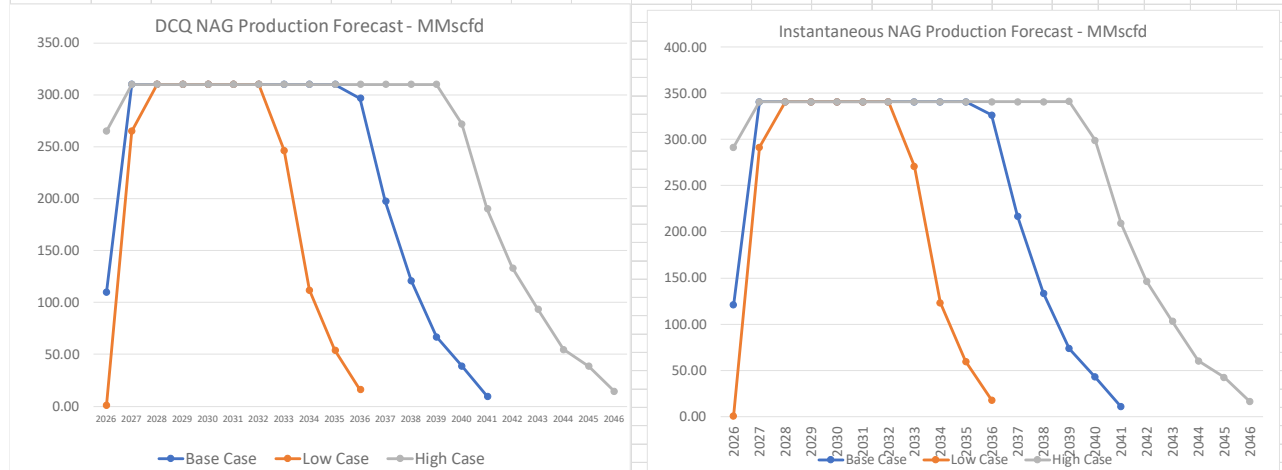
4.2. The FDP Production Forecast

The DCQ NAG Production Forecast from which the Instantaneous NAG Forecast is shown below, along with the Water Production Forecast.

The DCQ forecast includes 10% low and off, which when added back results on Instantaneous Production Forecast which is 110% of the DCQ forecast.

4.3. The Instantaneous Production Forecast

Gross DCQ NAG Production Forecast (MMscfd)																			
	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044
Base Case	109.77	310.00	310.00	310.00	310.00	310.00	310.00	310.00	310.01	310.03	296.74	197.31	121.16	66.78	38.90	9.50			
Low Case	0.63	264.99	310.00	310.00	310.00	310.01	310.00	246.41	111.79	53.94	15.96								
High Case	264.99	310.00	310.00	310.00	310.00	310.00	310.00	310.00	310.02	310.04	310.04	310.04	310.03	310.04	271.88	190.03	133.08	93.64	54.55



Gross Instantaneous NAG Production Forecast (MMscfd)																			
	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044
Base Case	120.75	341.00	341.00	341.00	341.00	341.00	341.00	341.00	341.01	341.03	326.41	217.04	133.27	73.46	42.79	10.45			
Low Case	0.69	291.49	341.00	341.00	341.00	341.01	341.00	271.05	122.97	59.34	17.55								
High Case	291.49	341.00	341.00	341.00	341.00	341.00	341.00	341.00	341.02	341.04	341.04	341.04	341.04	341.04	299.07	209.03	146.38	103.00	60.01

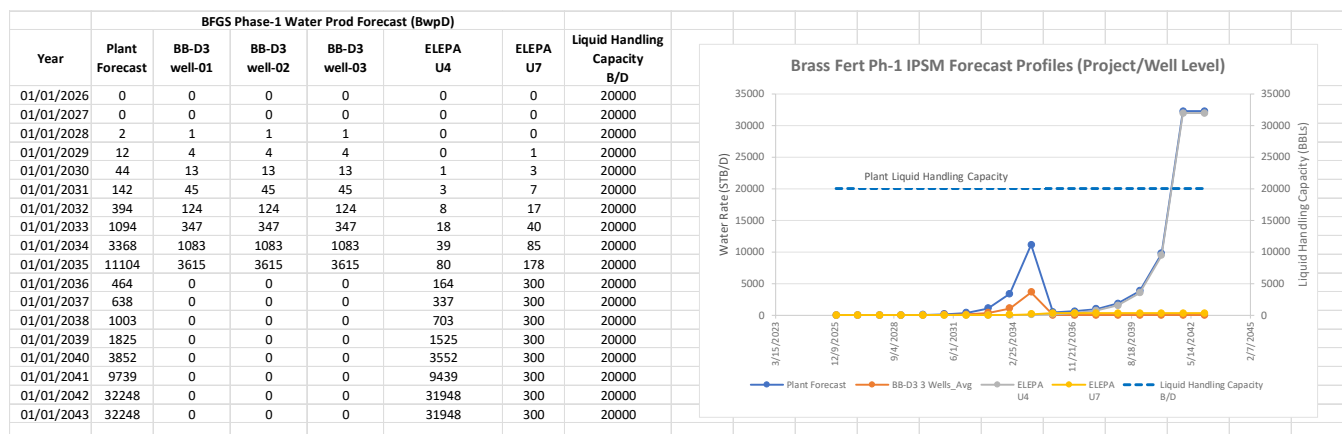


Figure 2-1: Satellite Image of the proposed BBOU – ELEP-AKAB Right of Way

The phase 2 pipelines routes are yet to be defined and surveyed.

4.4. The Flow Properties & Fluid Composition

BB-D3 Reservoir Fluid Composition & Properties					ELEP U4/U7 Reservoir Fluid Composition & Properties						
BB-D3 Reservoir Fluid Composition					ELEP U4/U7 averaged Res Fluid Composition						
					ELEP U4 Reservoir Fluid Properties						
Component		Flash Gas	Flash Oil	Res Fluid	Flash Gas		Flash Oil	Res Fluid			
No.		[mol %]	[mol %]	[mol %]	Density @ 60 [g/cm^3]		0.811				
1	N2	0.07	0.00	0.07	MW	[g/mol]	17.97	165.61	18.47		
2	CO2	0.59	0.00	0.59	GOR	[Scf/Stb]			180,012		
3	C1	93.26	0.00	92.92	Gravity	[air=1]	0.62				
4	C2	3.17	0.01	3.16	C7+	Mole %	0.45	97.62	0.53		
5	C3	1.01	0.03	1.01	C7+ MW	[g/mol]	100.89	167.69	143.17		
6	i-C4	0.61	0.07	0.61	C12+ MW	[g/mol]	-	191.6	191.6		
7	n-C4	0.44	0.07	0.44	CGR (bbbls/mmscf)				5.56		
8	i-C5	0.25	0.16	0.25							
9	n-C5	0.19	0.16	0.19							
10	C6	0.22	1.88	0.23							
11	C7	0.17	6.21	0.19							
12	C8	0.02	6.61	0.04							
13	C9	0.00	7.13	0.03							
14	C10	0.00	5.70	0.02							
15	C11	0.00	7.83	0.03							
16	C12+	0.00	64.14	0.22							
TOTAL		100	100	100							

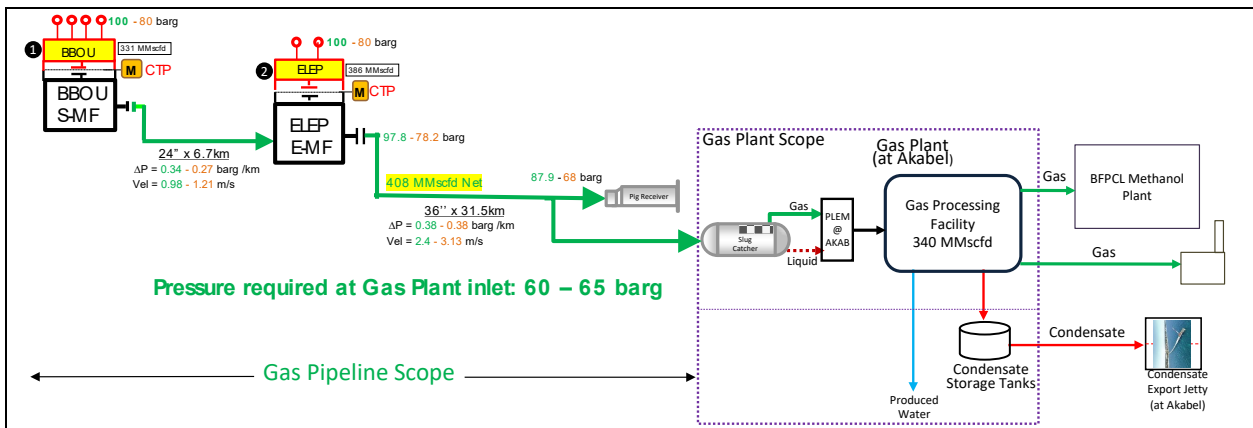
ELEP U4/U7 Reservoir Fluid Composition & Properties										
ELEP U4/U7 averaged Res Fluid Composition										
ELEP U4 Reservoir Fluid Properties										
Component		Flash Gas	Flash Oil	Res Fluid	Flash Gas		Flash Oil	Res Fluid		
No.		[mol %]	[mol %]	[mol %]	Density @ 60 [g/cm^3]		0.827			
1	N2	1.08	0.00	1.07	MW	[g/mol]	20.84	181.08	26.44	
2	CO2	1.15	0.00	1.11	GOR	[Scf/Stb]				
3	C1	85.97	0.00	84.17	Gravity	[air=1]	0.7190		0.9130	
4	C2	6.30	0.20	6.17	C7+	Mole %	1.13	94.98	4.41	
5	C3	2.37	0.26	2.32	C7+ MW	[g/mol]	105.67	186.72	166.70	
6	i-C4	0.64	0.27	0.64	C20+ MW	[g/mol]	-	339.50	339.50	
7	n-C4	0.66	0.40	0.65	CGR (bbbls/mmscf)					
8	i-C5	0.42	0.63	0.35						
9	n-C5	0.16	0.52	0.25						
10	C6	0.43	2.07	0.47	ELEP U7 Reservoir Fluid Properties					
11	C7	0.51	6.36	0.64			Flash Gas	Flash Oil	Res Fluid	
12	C8	0.33	13.14	0.59	Density @ 60 [g/cm^3]		0.839			
13	C9	0.00	12.10	0.24	MW	[g/mol]	19.43	175.14	20.55	
14	C10	0.00	9.33	0.19	GOR	[Scf/Stb]				
15	C11	0.00	7.05	0.14	Gravity	[air=1]	0.6710		0.7090	
16	C12	0.00	6.45	0.14	C7+	Mole %	0.54	96.34	1.24	
17	C13	0.00	6.58	0.14	C7+ MW	[g/mol]	105.66	179.07	146.96	
18	C14	0.00	6.09	0.13	C20+ MW	[g/mol]	-	338.80	338.80	
	C15	0.00	5.25	0.12	CGR (bbbls/mmscf)					
	C16	0.00	4.20	0.09						
	C17	0.00	3.18	0.07						
	C18	0.00	2.93	0.07						
	C19	0.00	1.93	0.04						
	C20+	0.00	11.10	0.25						
TOTAL		100.00	100.00	100.01						

5. Summary of Results

The results of the analysis are summarised below.

5.1. Phase-1 Facilities

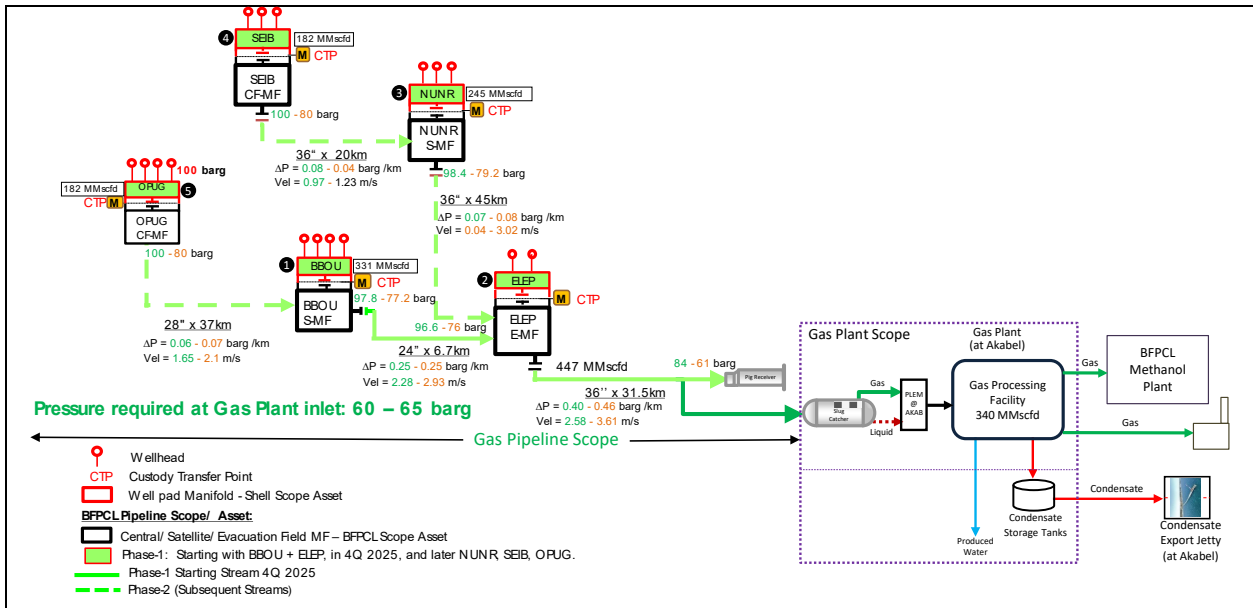
The result validates 24" for BBOU-ELEP, and 36" for ELEP-AKAB as shown below, satisfying the band of FTHPs.



Phase-1: PipeFlow Parameters: BBOU - ELEP - AKAB															
			Pipeline Size BBOU-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes BBOU & ELEP	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio (EVR)	
		S/N	in	in	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec		
Phase - 1	BBOU - ELEP	1	16	24	100	97.19		2.81	0.42	45.94	14.00	43.01	13.11	1.068	Remarks
					80	76.77		3.23	0.48	69.95	21.32	53.08	16.18	1.318	
		2	20	24	100	98.76		1.24	0.19	45.94	14.00	43.01	13.11	1.068	
					80	77.97		2.03	0.30	69.95	21.32	53.08	16.18	1.318	
		3	24	24	100	98.63		1.37	0.20	3.21	0.98	28.80	8.78	0.112	
					80	78.52		1.48	0.22	4.01	1.22	32.16	9.80	0.125	
		4	24	28	100	97.75		2.25	0.34	3.24	0.99	28.93	8.82	0.112	
					80	78.19		1.81	0.27	4.05	1.23	32.40	9.88	0.125	
		5	24	36	100	97.75		2.25	0.34	3.22	0.98	28.71	8.75	0.112	
					80	78.19		1.81	0.27	4.02	1.23	32.16	9.80	0.125	
	ELEP - AKAB	1	16	24	100	97.19	75.00	25.00	0.79	20.78	6.33	37.64	11.47	0.552	Meets Requirement
					80	76.77	46.00	34.00	1.08	34.46	10.50	48.47	14.77	0.711	Insufficient Pressure at AKAB
		2	20	24	100	98.76	77.00	23.00	0.73	20.22	6.16	37.17	11.33	0.544	Meets Requirement
					80	77.97	48.00	32.00	1.02	32.98	10.05	47.45	14.46	0.695	Insufficient Pressure at AKAB
		3	24	24	100	98.63	76.30	23.70	0.75	20.68	6.30	37.33	11.38	0.554	Meets Requirement
					80	78.52	48.00	32.00	1.02	33.42	10.19	47.45	14.46	0.704	Insufficient Pressure at AKAB
		4	24	28	100	97.75	84.66	15.34	0.49	13.39	4.08	35.42	10.80	0.378	Meets Requirement
					80	78.19	62.92	17.08	0.54	18.30	5.58	41.59	12.68	0.440	Meeting Requirement with Insufficient Margin
		5	24	36	100	97.75	87.92	12.08	0.38	7.85	2.39	34.73	10.59	0.226	Meets Requirement
					80	78.19	68.02	11.98	0.38	10.28	3.13	39.69	12.10	0.259	Meets Requirement
MAX TRIPUT	1	24	36	100	87.16	65	35.00	1.11	33.34	10.16	48.56	14.80	0.687	Based on Min Pressure Req'd at AKAB	
				80	77.09	65	15.00	0.48	17.90	5.45	44.94	13.70	0.398	Based on Min Pressure Req'd at AKAB	
MIN TRIPUT	2	24	36	100	97.74	65	35.00	1.11	3.47	1.06	27.86	8.49	0.125	Based on Min Vel for Pipe Cleaning	
				80	78.07	65	15.00	0.48	3.34	1.02	28.64	8.73	0.117	Based on Min Vel for Pipe Cleaning	

5.2. Phase-2 Facilities

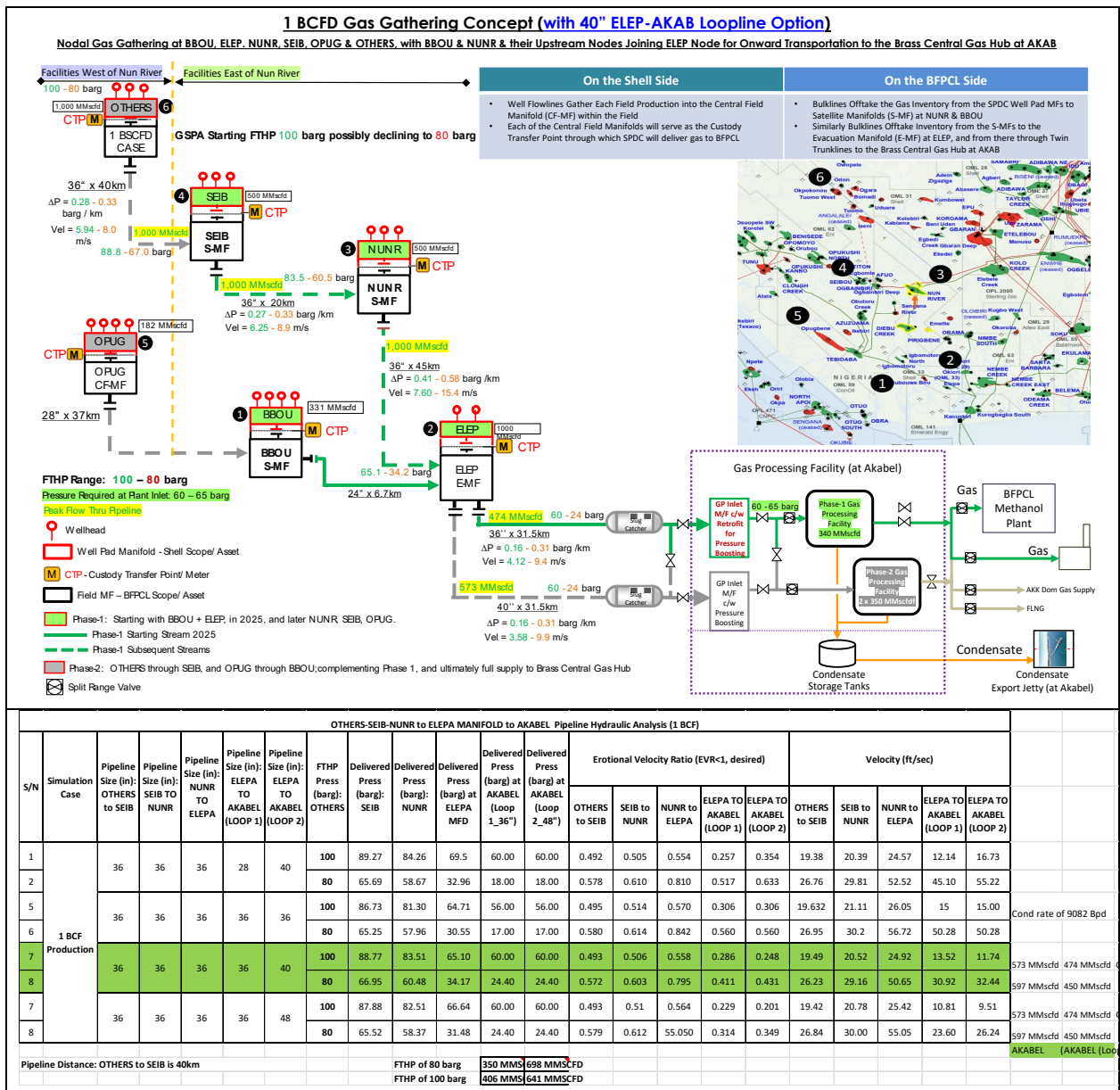
The result validates 24" for BBOU-ELEP, and 36" for ELEP-AKAB as shown below, satisfying the band of FTHPs.



Phase-2: PipeFlow Hydraulics Parameters: Total N nodes Flow to AKAB thru ELEP																					
		Pipeline Size ORUG-BBOU	Pipeline Size BBOU-ELEP	Pipeline Size SEB-NUNR	Pipeline Size NUNR-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes ORUG & SEB	Delivered Pressure BBOU M/F	Delivered Pressure NUNR M/F	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio (EVR)			
												Segment ELEP - AKAB									
		in	in	in	in	in	barg	barg	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec				
Phase-2	Segment ORUG - BBOU - ELEP	1	20	36	36	24	100	86.43		83.05											
							80	62.14		56.66											
		2	24	36	36	24	100	95.11		93.56											
							80	74.13		72.26											
		3	28	36	36	28	100	97.84		96.18											
							80	77.23		75.55											
	Segment SEB - NUNR - ELEP	4	28	36	36	32	100	98.54		97.02											
							80	77.40		75.59											
		5	28	36	36	36	100	98.14		96.54											
							80	78.46		76.10											
		6	24	36	36	36	100	95.17		93.55											
							80	74.14		72.39											
	Segment ELEP - AKAB	1	28	36	36	24	100		99.66	91.74		40.53	1.29	22.17	6.76	43.47	13.25	0.51			
							87		85.78	78.22		73.16	2.32	221.23	67.43	137.24	41.83	1.612			
		2	28	36	36	24	100		98.65	95.53		28.47	0.90	17.60	5.36	41.61	12.68	0.423			Meets Requirement
							82		80.4	78.16		40.55	1.29	34.70	10.58	58.32	17.78	0.595			Insufficient Pressure at AKAB
		3	28	36	36	28	100		99.22	95.47		18.46	0.59	12.04	3.67	38.47	11.72	0.313			Meets Requirement
							80		79	75.55		23.04	0.73	18.95	5.78	48.22	14.70	0.393			Insufficient Pressure at AKAB
	Segment ELEP - AKAB	4	28	36	36	32	100		99.59	96.46		12.56	0.40	8.48	2.58	37.03	11.29	0.229			Meets Requirement
							80		79.17	74.04		12.56	0.40	8.48	2.58	37.03	11.29	0.229			Insufficient Margin
		5	28	36	36	36	100		98.42	96.56		14.55	0.46	11.85	3.61	43.73	13.33	0.271			Meets Requirement
							80		79.17	75.55		14.55	0.46	11.85	3.61	43.73	13.33	0.271			Insufficient Margin
1		28	36	36	24	100		95.53	55.00		73.16	2.32	221.23	67.43	137.24	41.83	1.612				
						82		78.16	5.00		40.55	1.29	34.70	10.58	58.32	17.78	0.595				
Segment ELEP - AKAB	2	28	36	36	28	100		95.47	67.00		28.47	0.90	17.60	5.36	41.61	12.68	0.423			Meets Requirement	
						80		75.55	35.00		40.55	1.29	34.70	10.58	58.32	17.78	0.595			Insufficient Pressure at AKAB	
	3	28	36	36	32	100		96.46	78.00		18.46	0.59	12.04	3.67	38.47	11.72	0.313			Meets Requirement	
						80		74.04	51.00		23.04	0.73	18.95	5.78	48.22	14.70	0.393			Insufficient Pressure at AKAB	
	4	28	36	36	36	100		96.56	84.00		12.56	0.40	8.48	2.58	37.03	11.29	0.229			Meets Requirement	
						80		75.55	61.00		14.55	0.46	11.85	3.61	43.73	13.33	0.271			Insufficient Margin	
Segment ELEP - AKAB	5	24	36	36	36	100		96.56	84.00		12.56	0.40	8.48	2.58	37.03	11.29	0.229			Meets Requirement	
						80		75.55	61.00		14.55	0.46	11.85	3.61	43.73	13.33	0.271			Insufficient Margin	

5.3. The 1 BSCFD Case

The result validates 24" for BBOU-ELEP, and 36" for ELEP-AKAB, and 40" for the ELEP-AKAB Loopline as shown below, satisfying the band of FTHPs.



6. Design Basis

This section summarizes the assumptions, methodology and basis to obtain the various results during analysis of the Gas Gathering System. Based upon this analysis, a recommended pipeline system will be considered for FEED phase of design.

6.1. Simulation Scope

1. Carry out the pipeline network analysis of the entire gas production (phase 1 and phase 2) from different fields at their respective Flowing Tubing Head Pressures, anticipated pick flowrates, and minimum delivery pressure of 60 barg at Akabel.
2. To validate conceptual engineering recommended pipeline sizes for the anticipated peak flowrate from different fields.
3. Flow assurance checks (pressure drops and erosional velocity/fluid velocity/Erosional Velocity Ratio, EVR) for different scenarios of the pipeline network.
4. Document through the Pipeline Hydraulic Simulation Report the appropriate pipeline sizes and the operating conditions.

6.2. Production Profile

The pipeline system has been designed to achieve the production forecast detailed below. The production forecast includes the production from the OPUG, BBOU, SEIB, NUNR and ELEPA fields respectively, taken from the field manifolds after commingling the individual production wells.

Different production years were considered for each of the NAG producing field and for the maximum gas, maximum condensate and maximum water production at different scenarios of the simulation. This is highlighted in yellow in the summarized table 3.1 below.

The available manifold pressure to be provided by SPDC in line with the GSA is between 80 barg to 100 barg.

The actual required flowing tubing head pressure (FTHP) at manifold outlet will be determined by the simulation pipeline sizing sensitivity ensuring a delivery pressure range of 60 barg to 65 barg as explained in section 3.4.

The production profile as given in the tables 3.1 through 3.3 below were used for the hydraulic analysis.

Table 3.1: Production Profile Summary

BBOU PRODUCTION			
Year	Gas (MMSCFD)	Condensate (BCPD)	Water (BWPD)
7	305	1217	9002
9	331	1323	8038
18	174	16226	1667
ELEPA PRODUCTION			
14	386	22,905	16,513

BBOU PRODUCTION			
18	382	34,433	5,014
OPUG PRODUCTION			
16	154	9,082	3,859
18	174	16,226	1,667
20	182	15,606	2,775
SEIB PRODUCTION			
16	154	9,082	3,859
20	182	15,606	2,775
NUNR PRODUCTION			
13	245	14,583	10,844
OTHERS PRODUCTION			
	1,000	55000	3000

The detailed production forecast is also shown in Table 3.2 and 3.3 below.

Table 3.2: Detailed Production Forecast of BBOU Node (BBOU and OPUG)

BBOU TO ELEPA PIPELINE SEGMENT (24" x 6.4km)									
year	BBOU NODE			BUBOUWE BOU			OPUGBENE		
	TOTAL FLUID RATE								
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
0	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	181	944	907	181	944	907	0	0	0
2	181	878	907	181	878	907	0	0	0
3	181	788	907	181	788	907	0	0	0
4	181	742	7,855	181	742	7,855	0	0	0
5	181	707	7,547	181	707	7,547	0	0	0
6	215	840	8,491	215	840	8,491	0	0	0
7	305	1,217	9,002	305	1,217	9,002	0	0	0
8	327	1,323	8,193	327	1,323	8,193	0	0	0
9	331	1,338	8,038	331	1,338	8,038	0	0	0
10	93	1,763	2,527	67	239	2,397	26	1,525	129
11	75	3,242	1,301	21	59	1,031	54	3,183	270
12	74	3,457	1,074	16	42	784	58	3,416	290
13	71	4,084	1,513	2	4	85	69	4,080	1,429

	BBOU TO ELEPA PIPELINE SEGMENT (24" x 6.4km)										
14	71	4,161	3,048	0	0	0	71	4,161	3,048		
15	94	5,541	3,748	0	0	0	94	5,541	3,748		
16	154	9,082	3,859	0	0	0	154	9,082	3,859		
17	170	12,115	3,148	0	0	0	170	12,115	3,148		
18	174	16,226	1,667	0	0	0	174	16,226	1,667		
19	177	15,812	1,558	0	0	0	177	15,812	1,558		
20	182	15,606	2,775	0	0	0	182	15,606	2,775		

Table 3.3: Detailed Production Forecast of ELEPA Node

ELEPA TO AKABEL PIPELINE SEGMENT (36" x 31.5km)																							
ELEP NODE																							
year	TOTAL FLUID RATE				BUBOUWE BOU			ELEPA				NUN RIVER				SEIB				OPUG			
	MMscfd	bcpd	bwpd		MMscfd	bcpd	bwpd		MMscfd	bcpd	bwpd		MMscfd	bcpd	bwpd		MMscfd	bcpd	bwpd		MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)		(gas)	(cond.)	(water)		(gas)	(cond.)	(water)		(gas)	(cond.)	(water)		(gas)	(cond.)	(water)		(gas)	(cond.)	(water)
1	361	11,032	1,807		181	944	907		180	10,088	900		0	0	0		0	0	0		0	0	0
2	361	10,965	1,807		181	878	907		180	10,088	900		0	0	0		0	0	0		0	0	0
3	361	10,876	1,807		181	788	907		180	10,088	900		0	0	0		0	0	0		0	0	0
4	361	10,830	15,835		181	742	7,855		180	10,088	7,980		0	0	0		0	0	0		0	0	0
5	361	10,794	15,213		181	707	7,547		180	10,088	7,666		0	0	0		0	0	0		0	0	0
6	360	8,985	13,393		215	840	8,491		145	8,146	4,902		0	0	0		0	0	0		0	0	0
7	357	4,128	11,599		305	1,217	9,002		52	2,911	2,597		0	0	0		0	0	0		0	0	0
8	356	2,953	9,647		327	1,323	8,193		29	1,630	1,454		0	0	0		0	0	0		0	0	0
9	356	2,724	9,275		331	1,338	8,038		25	1,386	1,236		0	0	0		0	0	0		0	0	0
10	382	18,828	4,837		67	239	2,397		19	1,076	960		244	14,464	1,221		26	1,525	129		26	1,525	129
11	385	21,637	3,357		21	59	1,031		11	628	561		245	14,583	1,225		54	3,183	270		54	3,183	270
12	385	21,933	3,015		16	42	784		9	478	426		245	14,583	1,225		58	3,416	290		58	3,416	290
13	386	22,799	13,832		2	4	85		1	52	46		245	14,583	10,844		69	4,080	1,429		69	4,080	1,429
14	386	22,905	16,513		0	0	0		0	0	0		245	14,583	10,417		71	4,161	3,048		71	4,161	3,048
15	384	21,996	14,149		0	0	0		0	0	0		196	10,914	6,653		94	5,541	3,748		94	5,541	3,748
16	379	22,323	11,246		0	0	0		0	0	0		71	4,159	3,528		154	9,082	3,859		154	9,082	3,859
17	380	26,558	8,272		0	0	0		0	0	0		40	2,329	1,976		170	12,115	3,148		170	12,115	3,148
18	382	34,433	5,014		0	0	0		0	0	0		34	1,980	1,680		174	16,226	1,667		174	16,226	1,667
19	380	33,161	4,421		0	0	0		0	0	0		26	1,538	1,304		177	15,812	1,558		177	15,812	1,558
20	379	32,111	6,313		0	0	0		0	0	0		15	898	762		182	15,606	2,775		182	15,606	2,775

6.3. Fluid Composition

The fluid composition used for the simulation for the different field was extracted from the production and composition profile supplied by the project team. The well fluid compositions were provided on a dry basis. Different compositions for the fields are given below in fig 3.4.

Table 3.4: Fluid Composition

NODE	ELEPA TO AKABEL PIPELINE SEGMENT Fluid Composition (%)						BBOU TO ELEPA PIPELINE SEGMENT Fluid Composition (%)		
Components	BBOU	ELEP	NUNR	SEIB	OPUG	P/L Segment	BBOU	OPUG	P/L Segment
N ₂	0.07	NA	NA	NA	0.07	0.03514	0.07	0.07	0.0700
CO ₂	0.59	2.18	0.77	0.77	0.59	1.38185	0.59	0.59	0.5900
C ₁	92.92	84.165	72.83	72.83	92.92	88.5598	92.92	92.92	92.9200
C ₂	3.16	6.17	7.68	7.68	3.16	4.65904	3.16	3.16	3.1600
C ₃	1.01	2.32	4.85	4.85	1.01	1.66241	1.01	1.01	1.0100
iC ₄	0.61	0.635	1.46	1.46	0.61	0.62245	0.61	0.61	0.6100
nC ₄	0.44	0.65	1.97	1.97	0.44	0.54458	0.44	0.44	0.4400
iC ₅	0.25	0.35	0.95	0.95	0.25	0.2998	0.25	0.25	0.2500
nC ₅	0.19	0.245	0.76	0.76	0.19	0.21739	0.19	0.19	0.1900
C ₆	0.23	0.465	1.03	1.03	0.23	0.34703	0.23	0.23	0.2300
C ₇	0.19	0.635	1.41	1.41	0.19	0.41162	0.19	0.19	0.1900
C ₈	0.04	0.59	1.61	1.61	0.04	0.31391	0.04	0.04	0.0400
C ₉	0.03	0.24	0.9	0.9	0.03	0.13458	0.03	0.03	0.0300
C ₁₀	0.02	0.185	0.65	0.65	0.02	0.10217	0.02	0.02	0.0200
C ₁₁	0.03	0.14	0.47	0.47	0.03	0.08478	0.03	0.03	0.0300
C ₁₂₊	0.068	0.135	0.37	0.37	0.068	0.10137	0.07	0.07	0.0680
C ₁₃	0.047	0.14	0.34	0.34	0.047	0.09332	0.05	0.05	0.0470
C ₁₄	0.032	0.13	0.3	0.3	0.032	0.08081	0.03	0.03	0.0320
C ₁₅	0.022	0.115	0.26	0.26	0.022	0.06832	0.02	0.02	0.0220
C ₁₆	0.015	0.09	0.21	0.21	0.015	0.05235	0.02	0.02	0.0150
C ₁₇	0.01	0.07	0.16	0.16	0.01	0.03988	0.01	0.01	0.0100
C ₁₈	0.007	0.065	0.14	0.14	0.007	0.03589	0.01	0.01	0.0070
C ₁₉	0.016	0.04	0.08	0.08	0.016	0.02795	0.02	0.02	0.0160
C ₂₀₊	NA	0.25	0.8	0.8	NA	0.12451	NA	NA	NA
Total	100.00	100.01	100.00	100.00	100.00	100.00	100.00	100.00	100.00

6.4. Ambient Conditions

The following were considered as the ambient condition:

Table 3.5: Ambient Conditions

	Units	
Minimum ambient temperature	°C	18
Minimum Soil temperature	°C	23
Minimum swamp temperature	°C	23

Manifold Conditions

The manifold conditions for the various fields are presented in Table 3.6 below. This is based on current available data. The lowest manifold temperature of 45°C is used for the manifold temperature as the inlet to the pipeline system.

Table 3.6: Manifold Conditions

S/N	Field	Manifold Flowing Temperature (°C)	Manifold Flowing Pressure (barg)
1	BBOU	45 – 95	80 – 100
2	ELEPA	45 – 95	80 – 100
3	OPUG	45 – 95	80 – 100
4	SEIB	45 – 95	80 – 100
5	NUNR	45 – 95	80 – 100

6.5. Pipeline Delivery Pressure

The gas from different fields will delivered to the Gas Plant at Akabel. The pipeline system gas delivery pressure is between 60 barg to 65 barg. The delivery pressure is the inlet pressure requirement at Akabel Gas Plant.

6.6. Pipeline Profile

The pipeline topography is as shown in the graph fig 3.1 and fig 3.2 below. The pipeline burial depth is assumed to be 1.5 meter below the mean sea level. The wellhead platform for the producing fields is assumed to be elevated (5m above the mean sea level).

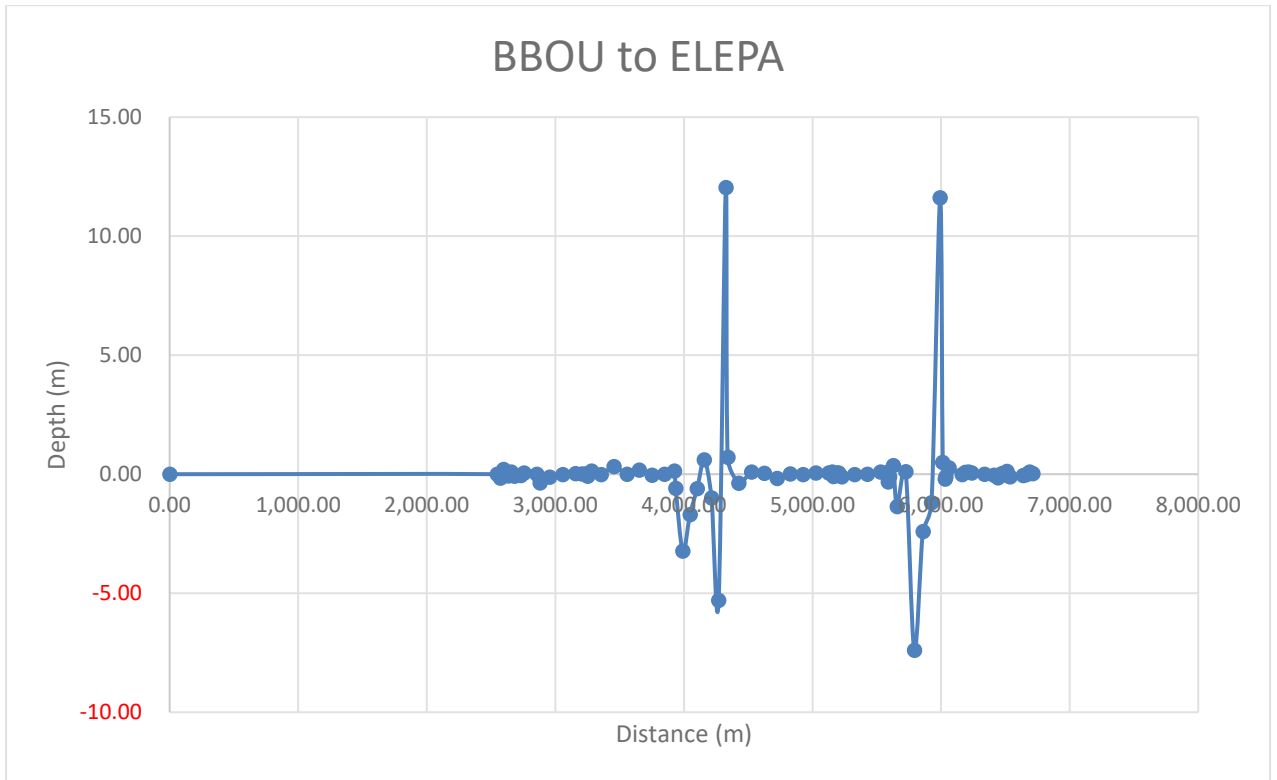


Fig 3.1: Pipeline profile from BBOU to ELEPA

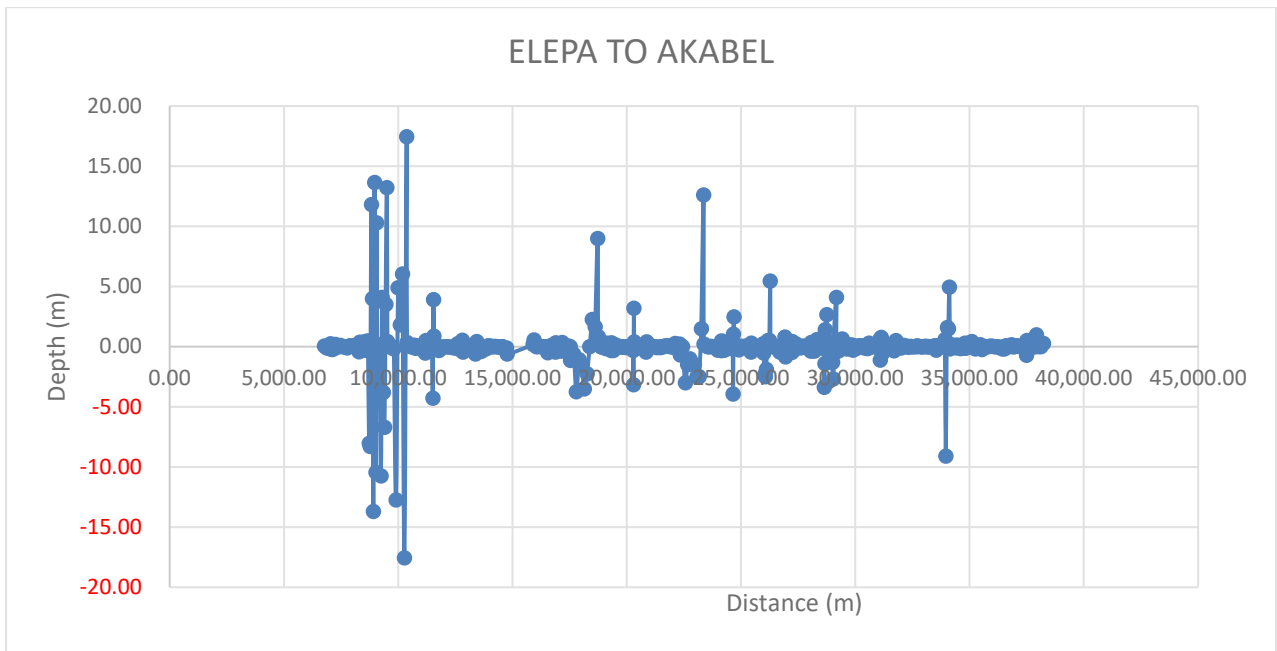


Fig 3.2: Pipeline profile from ELEPA to AKABEL

6.7. Pipeline Characteristics

The pipeline wall thickness used for the analysis are presented below in the table below:

Table 3.7: pipeline characteristics

Wall Section	Material	Outer Diameter mm (in)	Thickness mm (in)	Inside Diameter mm (in)
Pipeline	24 - inch Carbon Steel	610 (24)	15.88 (0.625197)	577.85 (22.75)
	28 - inch Carbon Steel	711 (28)	17.48 (0.688189)	676.15 (26.62)
	36 - inch Carbon Steel	914 (36)	23.83 (0.938189)	866.65 (34.12)

A roughness of 0.0018in is used for the analysis.

6.8. Pipeline Sizing Criteria

The following aspects has been considered while sizing the pipeline from OPUG to BBOU, BBOU to ELEPA, ELEPA to AKABEL and from SEIB to NUNR to ELEPA to AKABEL.

1. System deliverability, this includes delivering a minimum of 60 barg at AKABEL with an inlet pressure/FTHP range of 80 barg to 100 barg. This is applicable to all scenarios of the pipeline network to AKABEL.
2. Erosional Velocity Ratio (EVR) of < 1 , to avoid the erosion of the pipeline. EVR is a ratio of fluid velocity to the erosional velocity calculated from the erosion model, API 14E.

7. Approach to Work

This section describes the process simulation structure and methodology. The process description is however based on the gas production scenarios coming differently to supply gas to AKABEL as developed during the conceptual stage of this project and aligning with the pipeline route survey map/report.

7.1. Simulation Tools

Version 2017.1 of PIPESIM Hydraulics software was used in performing the hydraulics simulation of the pipelines systems. UNISIM R460.1 was used to create the gas, condensate and water compositions from the available bulk fluid compositions after saturation with water.

7.2. Steady State Simulations

For the hydraulic analysis simulations, the correlation for both vertical and horizontal Pipes was selected as Hagedorn & Brown and Beggs & Bill Revised respectively. The Table 4.1 below shows the thermodynamics simulation set up data for the Pipesim simulation set up.

Table 4.1 – Pipesim Simulation Set-up

Simulation Set up	Method
Simulation Type	Network Model
Fluid Type	Compositional
Property Package (Equation of State-EOS)	Peng Robinson

3-Phase Composition: Since the available composition was the bulk fluid composition from the manifold and the available production profile/flowrate was the gas, condensate and water flowrate, UNISIM R460.1 simulation tool was used to create the gas, condensate and water compositions for proper data input and analysis

The bulk composition, which was supplied on dry basis, was saturated with water and passed through a three-phase separator operating at a 100 barg pressure for both BBOU Node and ELEPA Node to generate the phase composition needed to represent the gas, condensate and water productions, fig 4.1 and 4.2.

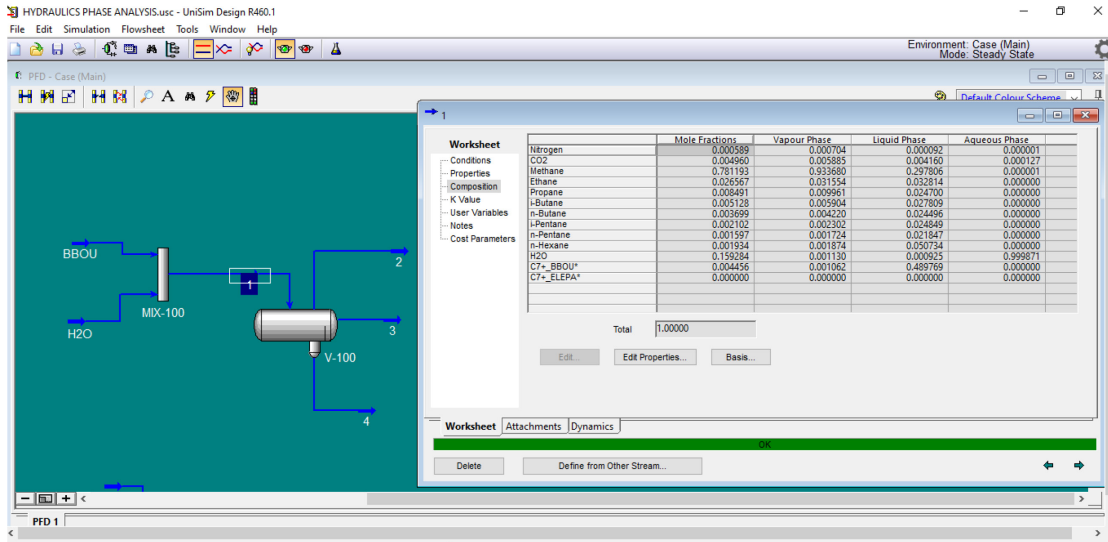


Fig 4.1: BBOU Node (BBOU/OPUG) 3-Phase Composition

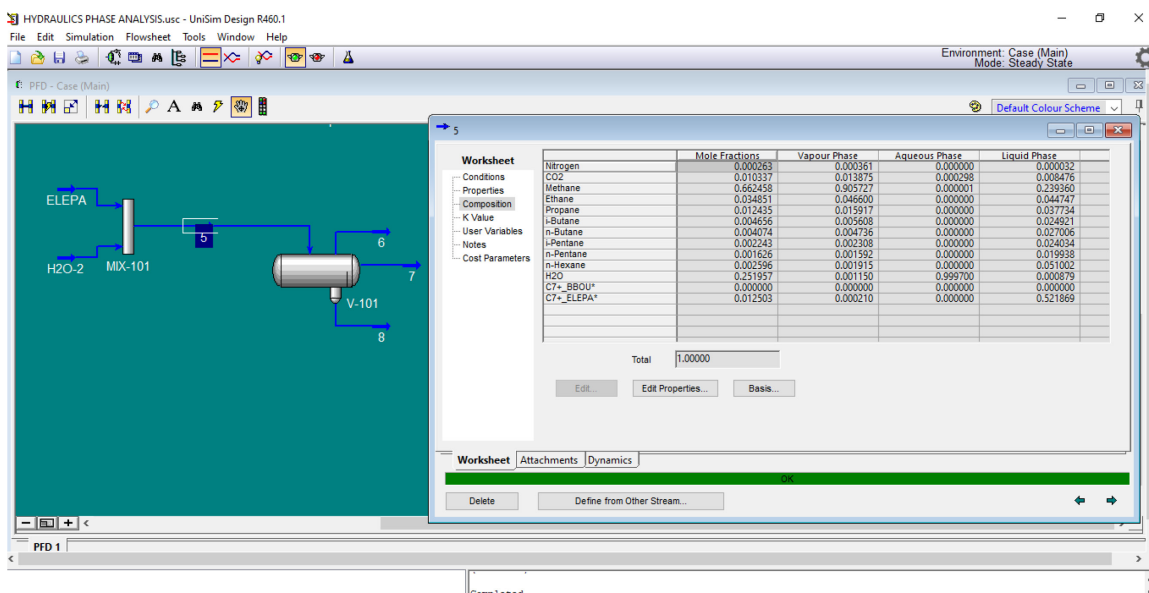


Fig 4.2: ELEPA Node (ELEPA/SEIB/NUNR) 3-Phase Composition

7.3. Simulation Scenarios

Different productions were considered from different fields, as shown in the Gas Gathering Pipeline Schematics fig 4.3. Each of the production scenarios from either of the fields; OPUG, BBOU, SEIB, NUNR and ELEPA (or a combination of two or more) were considered at different Flowing Tubing Head Pressure (FTHP) ranging from 100 barg to 80 barg in order to deliver at a pressure range of 60barg to 65barg to Akabel. At a minimum FTHP of 80 barg from the field manifolds, different pipeline sizes were considered.

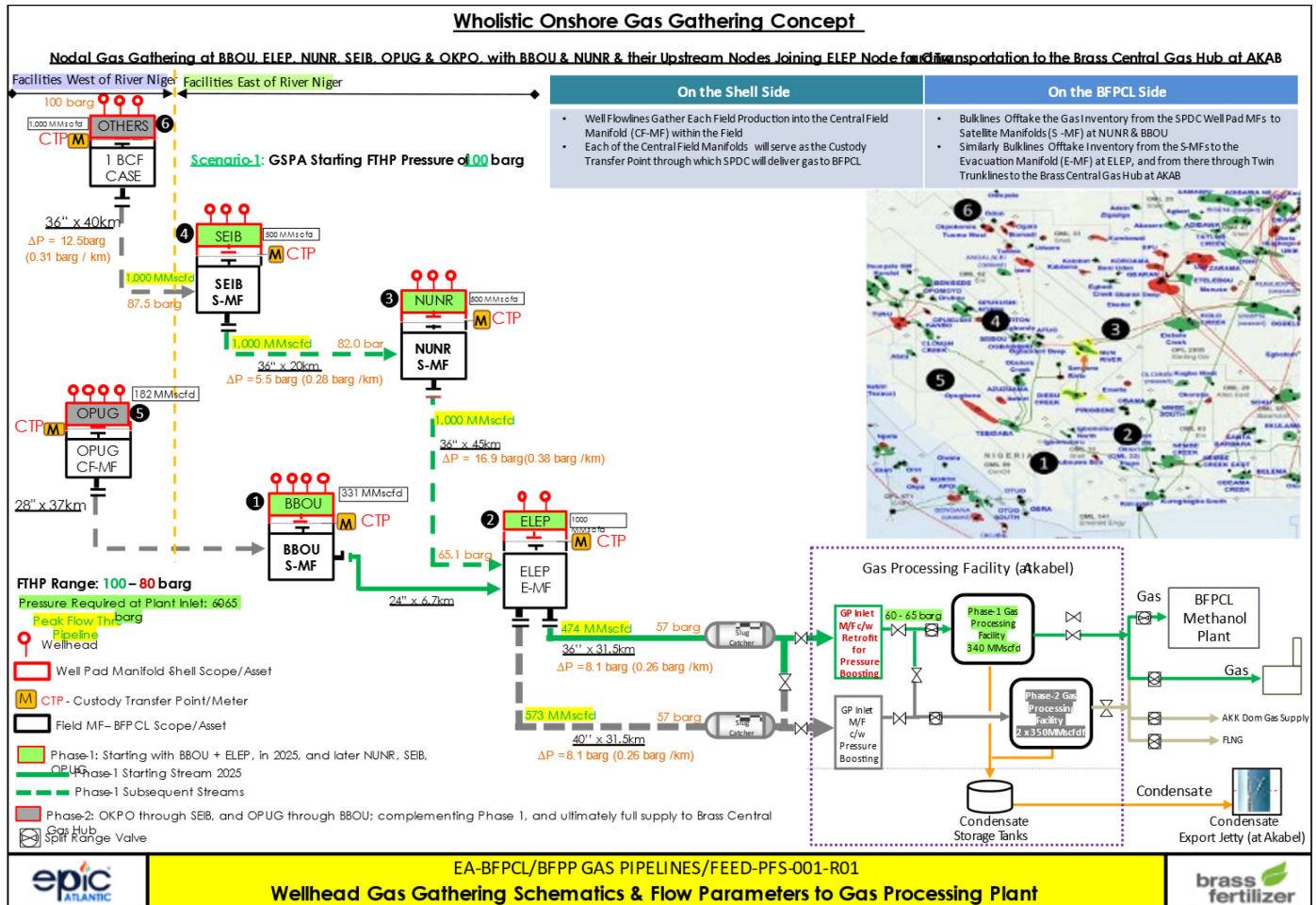


Fig 4.3: Gas Gathering Pipeline Schematics

7.3.1. Phase 1: BBOU to ELEPA Scenario

This section of the phase 1 involves the production from the BBOU to ELEPA. The production considered maximum rate of 331 MMSCFD, 16,200 BCPD and 9,000 BWPD from the BBOU and different Flowing Tubing Head Pressure (FTHP) ranging from 100 barg to 80 barg. This production form the basis for hydraulic analysis of the pipeline from BBOU to ELEPA, which is 6.7km apart. The simulation snapshot is shown in fig 4.4.

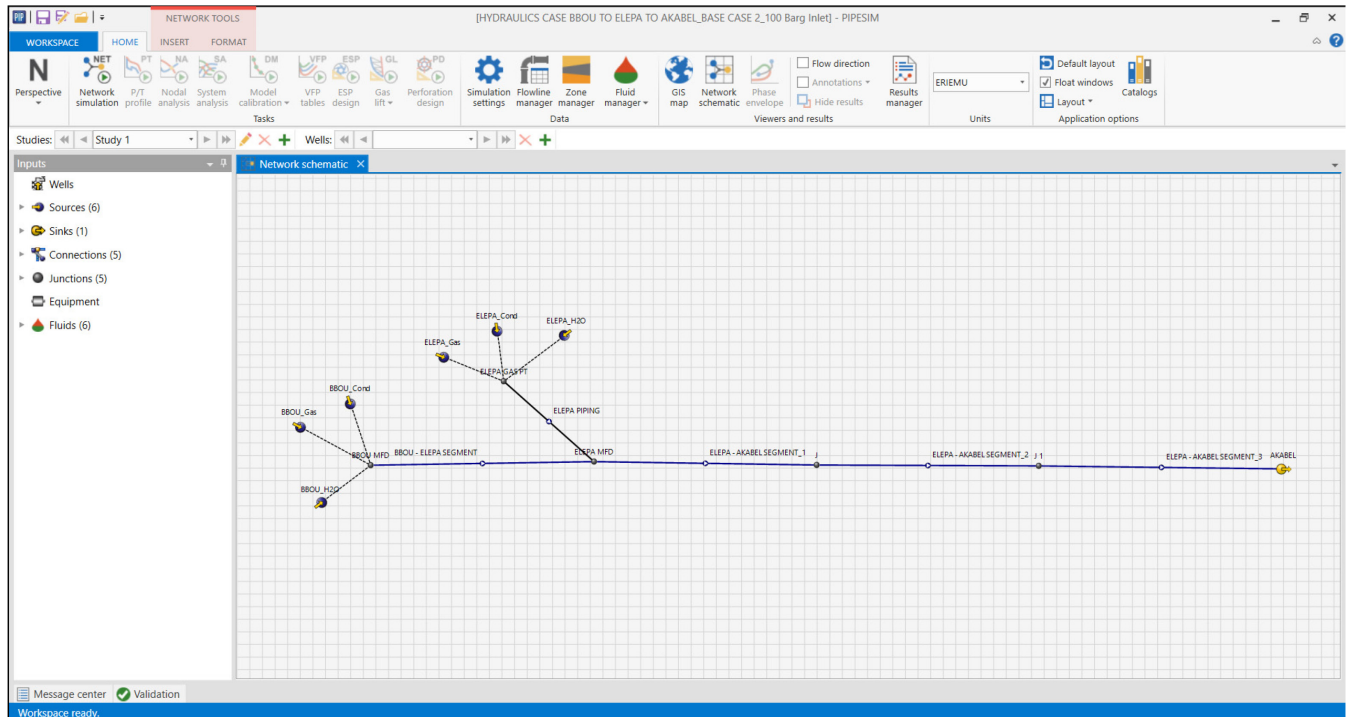


fig 4.4: BBOU to ELEPA Hydraulics Simulation Snapshot.

7.3.2. Phase 1: ELEPA to AKABEL Scenario

Here, the production from the ELEPA to AKABEL at a distance of 31.5km were considered. The production considered at ELEPA were a combination of productions from different fields as per the production year and combined at ELEPA. The combined flow is expected to have enough pressure to arrive at AKABEL at a minimum pressure of 60 barg. A combined production rate of 386 MMSCFD, 34.4k BCPD and 9k BWPD was expected to flow from the ELEPA to AKABEL.

7.3.3. Phase 2: OPUG to BBOU to ELEPA Segment

This scenario involves the production from OPUG to BBOU to ELEPA. The peak production rate of 182 MMSCFD, 16.2k BCPD and 3.9k BWPD from OPUG as shown in Fig 4.5 was used for the hydraulic analysis of the 37km pipeline from OPUG to BBOU. Different pipeline combinations were considered. This scenario forms the basis for the sizing the OPUG to BBOU pipeline.

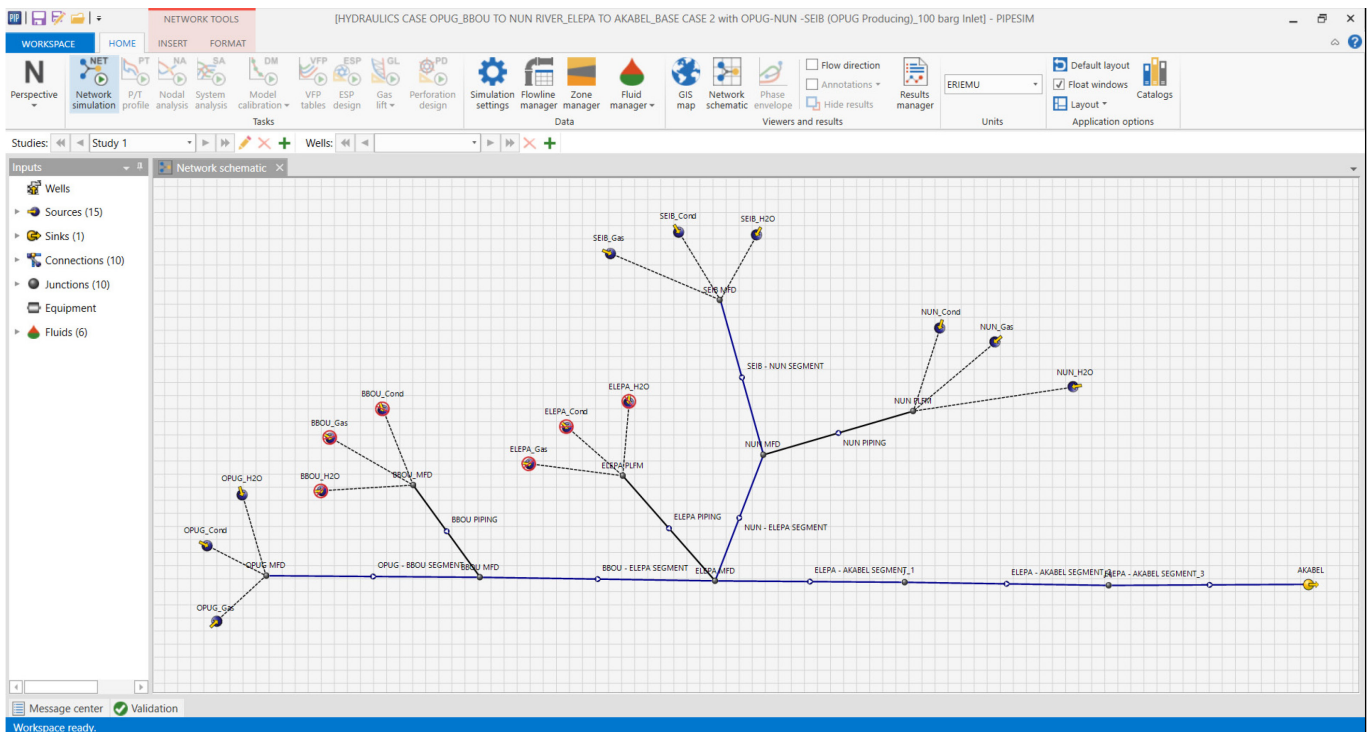


Fig 4.5: OPUG to BBOU to ELEPA Hydraulics Simulation Snapshot.

7.3.4. Phase 2: SEIB to NUNR to ELEPA Segment

This production scenarios covers the SEIB peak production of 182 MMSCFD, 16.2k BCPD and 3.9k BWPD and NUNR peak production of 245 MMSCFD, 14.6k BCPD and 10.8k BWPD. The field is 20km to the NUNR and the NUNR is 45km to ELEPA. Different scenarios were considered to size the pipeline sizes from SEIB to NUNR to ELEPA. Simulation snapshot shown in fig 4.6.

7.3.5. Phase 2: ELEPA to AKABEL Segment

This scenario combines flowrate from the SEIB and NUNR through ELEPA to AKABEL. The SEIB peak production of 182 MMSCFD, 16.2k BCPD and 3.9k BWPD and NUNR peak production of 245 MMSCFD, 14.6k BCPD and 10.8k BWPD were used for the simulation. The ELEPA field is 31.5km to AKABEL. Simulation snapshot shown in fig 4.6.

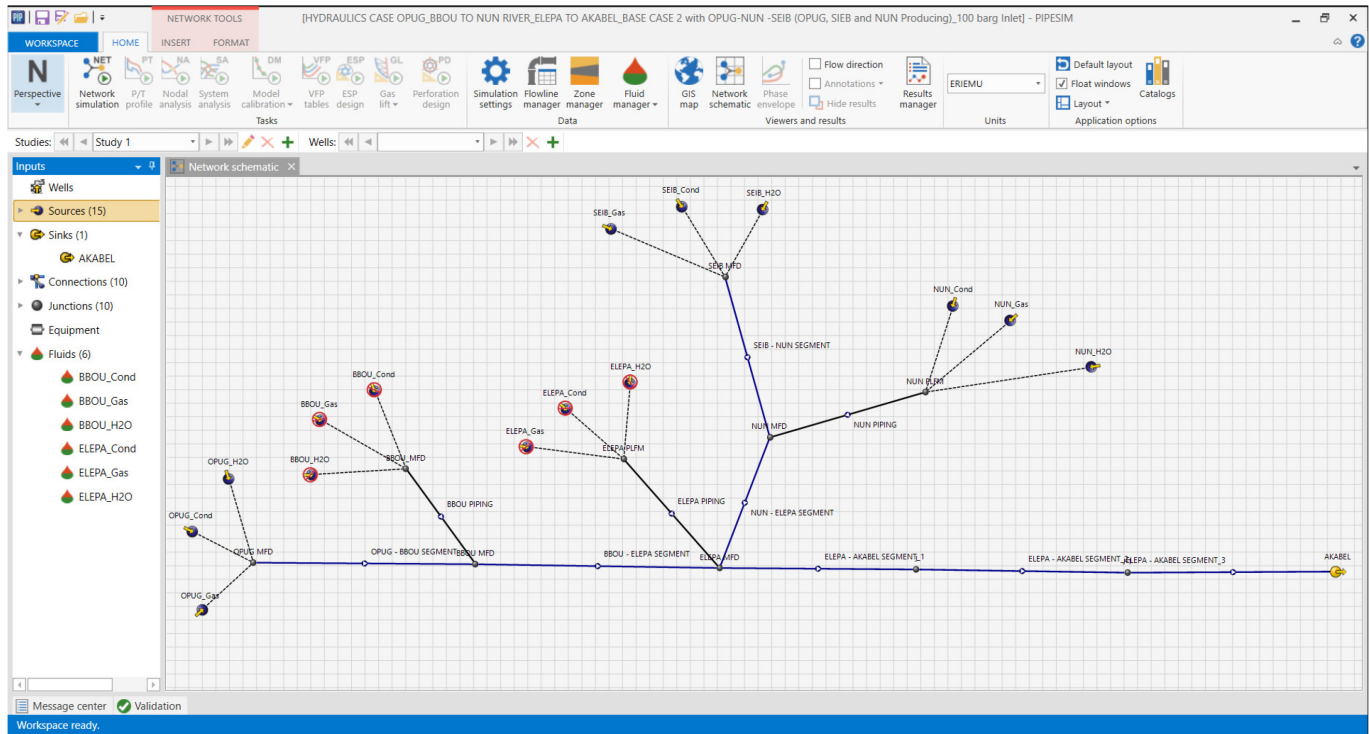
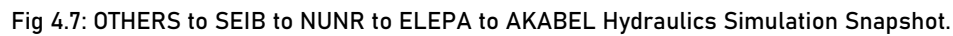


Fig 4.6: SEIB to NUNR to ELEPA to AKABEL Hydraulics Simulation Snapshot.

7.3.6. OTHERS to SEIB to NUNR to ELEPA to AKABEL Scenario

Due to the requirement for the future production of 1,000 MMSCFD (1BCFPD) from Others/SEIB/NUNR fields, considerations were given to this production to AKABEL, with an inlet pressure range of 80 barg to 100 barg. The peak production rate of 1,000 MMSCFD, 55k BCPD and 3k BWPD from OTHERS field as shown in Fig 4.7. The 2-loop production systems from ELEPA to AKABEL were used to deliver the bulk flow from ELEPA to AKABEL because the 36" ELEPA to AKABEL pipeline would have been existing prior to the commencement of the 1 BCF production, thus there is need for a second line to ensure adequate delivery to AKABEL.



8. Pipesim Hydraulics Simulation Results

The result of the hydraulic analysis performed at the two different Manifold Inlet Flowing Tubing Head Pressure (FTHP) conditions of 100 barg and 80 barg and a minimum desired delivery pressure of 60 barg at AKABEL (except for 1 BCFD production only when the desired delivery pressure of 5 barg to 10 barg is required at AKABEL).

8.1. Phase 1: BBOU to ELEPA Segment

This scenario shows the deliverability of the peak/maximum flowrates of gas, condensate and water (331 MMSCFD, 16,200 BCPD and 9,000 BWPD) from BBOU to ELEPA through different pipeline sizes as per the Sensitivity analysis. The simulation result is as shown in table 5.1 below.

Phase-1: PipeFlow Parameters: BBOU - ELEP - AKAB														
			Pipeline Size BBOU-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes BBOU & ELEP	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio
		S/N	in	in	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec	
Phase - 1	BBOU - ELEP	1	16	24	80	76.77		3.23	0.48	69.95	21.32	53.08	16.18	1.318
					100	97.19		2.81	0.42	45.94	14.00	43.01	13.11	1.068
		2	20	24	80	77.97		2.03	0.30	69.95	21.32	53.08	16.18	1.318
					100	98.76		1.24	0.19	45.94	14.00	43.01	13.11	1.068
		3	24	24	80	78.52		1.48	0.22	4.01	1.22	32.16	9.80	0.125
					100	98.63		1.37	0.20	3.21	0.98	28.80	8.78	0.112
		4	24	28	80	78.19		1.81	0.27	4.05	1.23	32.40	9.88	0.125
					100	97.75		2.25	0.34	3.24	0.99	28.93	8.82	0.112
		5	24	36	80	78.19		1.81	0.27	4.02	1.23	32.16	9.80	0.125
					100	97.75		2.25	0.34	3.22	0.98	28.71	8.75	0.112

Fig 5.1: Sensitivity Analysis Result of BBOU – ELEPA Hydraulic Analysis.

From the above table, 24" pipeline size from the BBOU to ELEPA will be adequate as it does not violate the Erosional Velocity Ratio criteria of < 1 .

8.2. Phase 1: ELEPA to AKABEL Segment

This scenario shows the deliverability of a combined peak production rate of gas 386 MMSCFD, condensate 34.4k BCPD and 9k BWPD water from ELEPA to AKABEL through different pipeline sizes as in the Sensitivity analysis. The simulation result is as shown in table 5.2 below.

Phase-1: PipeFlow Parameters: ELEP - AKAB														
			Pipeline Size BBOU-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes BBOU & ELEP	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio (EVR)
		S/N	in	in	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec	
Phase - 1	ELEP - AKAB	1	16	24	80	76.77	46.00	34.00	1.08	34.46	10.50	48.47	14.77	0.711
					100	97.19	75.00	25.00	0.79	20.78	6.33	37.64	11.47	0.552
		2	20	24	80	77.97	48.00	32.00	1.02	32.98	10.05	47.45	14.46	0.695
					100	98.76	77.00	23.00	0.73	20.22	6.16	37.17	11.33	0.544
		3	24	24	80	78.52	48.00	32.00	1.02	33.42	10.19	47.45	14.46	0.704
					100	98.63	76.30	23.70	0.75	20.68	6.30	37.33	11.38	0.554
		4	24	28	80	78.19	62.92	17.08	0.54	18.3	5.58	41.59	12.68	0.44
					100	97.75	84.66	15.34	0.49	13.39	4.08	35.42	10.80	0.378
		5	24	36	80	78.19	68.02	11.98	0.38	10.28	3.13	39.69	12.10	0.259
					100	97.75	87.92	12.08	0.38	7.85	2.39	34.73	10.59	0.226

Fig 5.2: Sensitivity Analysis Result of ELEPA – AKABEL Hydraulic Analysis.

From the above table, 28" and 36" pipeline size from the ELEPA to AKABEL will be adequate as it does not violate the Erosional Velocity Ratio criteria of < 1 and successfully delivers pressure greater than or equal to 60 barg at AKABEL.

8.3. Phase 2: OPUG to BBOU to ELEPA Pipeline Network Segment

An integrated sensitivity analysis was performed at different pressure scenario as shown in result table 5.3 below to ascertain the adequate pipeline size from OPUG to BBOU with an integrated effect of delivering a minimum of 60barg at AKABEL.

The hydraulics of OPUG to BBOU successfully favoured the 28" pipeline to adequately deliver a minimum of 76 barg at ELEPA, even though all the pipelines sizes in the sensitivity never violated EVR criteria, as shown in result table 5.3 below. Pressure of 76 barg will be good enough to deliver a minimum of 60 barg at AKABEL.

Phase-2: PipeFlow Hydraulics Parameters: Total Nodes Flow to AKAB thru ELEP																												
			Pipeline Size OPUG-BBOU	Pipeline Size BBOU-ELEP	Pipeline Size SEIB-NUNR	Pipeline Size NUNR-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes OPUG & SEIB	Delivered Pressure BBOU M/F	Delivered Pressure NUNR M/F	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio		
													Segment OPUG - BBOU								Segment BBOU - ELEP							
			in	in	in	in	in	barg	barg	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec		barg	barg / km	ft / sec	m / sec	ft / sec	m / sec			
Phase-2	Segment OPUG - BBOU - ELEP	1	20	20	36	36	24	80	62.14		56.66		17.86	0.48	19.08	5.82	43.85	13.37	0.435	5.48	0.82	20.51	6.25	45.47	13.86	0.451		
								100	86.43		83.05		13.57	0.37	13.45	4.10	36.81	11.22	0.365	3.38	0.50	14.01	4.27	37.58	11.45	0.373		
		2	24	24	36	36	24	80	74.13		72.26		5.87	0.16	9.73	2.97	39.71	12.10	0.245	1.87	0.28	10.08	3.07	40.48	12.34	0.249		
								100	95.11		93.56		4.89	0.13	7.60	2.32	35.02	10.68	0.217	1.55	0.23	7.71	2.35	35.37	10.78	0.218		
		3	28	24	36	36	28	80	77.23		75.55		2.77	0.07	6.88	2.10	38.08	11.91	0.176	1.68	0.25	9.62	2.93	39.49	12.04	0.244		
								100	97.84		96.18		2.16	0.06	5.40	1.65	34.63	10.55	0.156	1.66	0.25	7.46	2.28	34.79	10.60	0.215		
		4	28	24	36	36	32	80	77.40		75.59		2.60	0.07	6.86	2.09	39.01	11.89	0.176	1.81	0.27	9.60	2.93	39.45	12.02	0.243		
								100	98.54		97.02		1.46	0.04	5.37	1.64	34.51	10.52	0.155	1.52	0.23	7.41	2.26	34.66	10.56	0.214		
		5	28	24	36	36	36	80	78.46		76.10		1.54	0.04	6.82	2.08	38.97	11.88	0.175	2.36	0.35	9.50	2.90	39.26	11.97	0.242		
								100	98.14		96.54		1.86	0.05	5.42	1.65	34.74	10.59	0.156	1.60	0.24	7.48	2.28	34.79	10.60	0.215		
		6	24	24	36	36	36	80	74.14		72.39		5.86	0.16	9.78	2.98	39.76	12.12	0.246	1.75	0.26	10.02	3.05	40.24	12.27	0.249		
								100	95.17		93.55		4.83	0.13	7.70	2.35	35.32	10.77	0.218	1.62	0.24	7.79	2.37	35.57	10.84	0.219		

Fig 5.3: Sensitivity Analysis Result of OPUG – BBOU – ELEPA Hydraulic Analysis.

8.4. Phase 2: SEIB to NUNR to ELEPA Pipeline Network Segment

At a certain time in the life of the plant, production from the other fields would have been depleted and thus there is need for hydraulic analysis of pipeline producing from the SEIB to NUNR to ELEPA. The sensitivity analysis was performed using different pipeline arrangement from SEIB to NUNR to ELEPA at different pressure scenario as shown in result table 5.4 below.

The pipeline sensitivity analysis also confirmed the 28" SEIB to NUNR and 28" pipeline from NUNR to ELEPA will deliver the peak volumes to AKABEL via ELEPA and also obeys the flow assurance criteria of $EVR < 1$. However, the arrival pressure at ELEPA will not be able to deliver the fluid to AKABEL at a minimum pressure of 60 barg.

Further sensitivity of the pipeline network for the pick production volumes coming from the SEIB to NUNR to ELEPA, confirmed that 36" pipeline from SEIB to NUNR and 36" pipeline from NUNR to ELEPA will be adequate for delivering the production at AKABEL at minimum pressure of 60 barg while obeying the velocity/erosional velocity criteria.

Phase-2: PipeFlow Hydraulics Parameters: Total Nodes Flow to AKAB thru ELEP																												
			Pipeline Size OPUG-BBOU	Pipeline Size BBOU-ELEP	Pipeline Size SEIB-NUNR	Pipeline Size NUNR-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes OPUG & SEIB	Delivered Pressure BBOU MF	Delivered Pressure NUNR MF	Delivered Pressure ELEP MF	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio (EVR)	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio (EVR)		
													Segment SEIB - NUNR								Segment NUNR - ELEP							
													barg	barg / km	ft / sec	m / sec	ft / sec	m / sec		barg	barg / km	ft / sec	m / sec	ft / sec	m / sec			
Phase-2	Segment SEIB - NUNR - ELEP	1	28	24	28	28	24	87		85.78	78.22		1.22	0.06	2.54	0.77	32.56	9.93	0.078	7.56	0.17	11.85	3.61	37.15	11.32	0.319		
								100		99.66	91.74		0.34	0.02	2.17	0.66	30.14	9.19	0.07	7.92	0.18	9.87	3.01	33.92	10.34	0.291		
		2	28	24	36	36	24	82		80.40	78.16		1.60	0.08	1.63	0.50	33.90	10.33	0.048	2.24	0.05	7.205	2.20	37.14	11.32	0.194		
								100		98.65	95.53		1.35	0.07	1.32	0.40	30.00	9.14	0.044	3.12	0.07	5.78	1.76	33.22	10.12	0.174		
		3	28	24	36	36	28	80		79.00	75.55		1.00	0.05	4.02	1.23	40.61	12.38	0.099	3.45	0.08	9.84	3.00	39.84	12.14	0.247		
								100		99.22	95.47		0.78	0.04	3.17	0.97	36.00	10.97	0.088	3.75	0.08	7.66	2.33	35.14	10.71	0.218		
		4	28	24	36	36	32	80		79.17	74.04		0.83	0.04	4.04	1.23	40.40	12.31	0.100	5.13	0.11	9.95	3.03	40.12	12.23	0.248		
								100		99.59	96.46		0.41	0.02	3.16	0.96	35.91	10.95	0.088	3.13	0.07	7.63	2.33	35.00	10.67	0.218		
		5	28	24	36	36	36	80		79.17	75.55		0.83	0.04	4.03	1.23	40.30	12.28	0.100	3.62	0.08	9.91	3.02	39.96	12.18	0.248		
								100		98.42	96.56		1.58	0.08	3.17	0.97	36.02	10.98	0.088	1.86	0.04	7.69	2.34	35.28	10.75	0.218		

Fig 5.4: Sensitivity Analysis Result of SEIB – NUNR – ELEPA Hydraulic Analysis.

8.5. Phase 2: ELEPA to AKABEL Pipeline Network Segment

The peak gas, condensate and water production from SEIB and NUNR to ELEPA shows deliverability to AKABEL via a 36" pipeline as confirmed by the sensitivity analysis shown in fig 5.5.

Phase-2: PipeFlow Hydraulics Parameters: Total Nodes Flow to AKAB thru ELEP																				
			Pipeline Size OPUG-BBOU	Pipeline Size BBOU-ELEP	Pipeline Size SEIB-NUNR	Pipeline Size NUNR-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes OPUG & SEIB	Delivered Pressure BBOU M/F	Delivered Pressure NUNR M/F	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop		Velocity		Erosional Velocity		Erosional Velocity Ratio (EVR)	
													Segment ELEP - AKAB							
			in	in	in	in	in	barg	barg	barg	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec	
Phase-2	Segment ELEP - AKAB	1	28	24	36	36	24	82			78.16	5.00	73.16	2.32	221.23	67.43	137.24	41.83	1.612	
								100			95.53	55.00	40.53	1.29	22.17	6.76	43.47	13.25	0.510	
		2	28	24	36	36	28	80			75.55	35.00	40.55	1.29	34.70	10.58	58.32	17.78	0.595	
								100			95.47	67.00	28.47	0.90	17.60	5.36	41.61	12.68	0.423	
		3	28	24	36	36	32	80			74.04	51.00	23.04	0.73	18.95	5.78	48.22	14.70	0.393	
								100			96.46	78.00	18.46	0.59	12.04	3.67	38.47	11.72	0.313	
		4	28	24	36	36	36	80			75.55	61.00	14.55	0.46	11.85	3.61	43.73	13.33	0.271	
								100			96.56	84.00	12.56	0.40	8.48	2.58	37.03	11.29	0.229	
		5	24	24	36	36	36	80			75.55	61.00	14.55	0.46	11.85	3.61	43.73	13.33	0.271	
								100			96.56	84.00	12.56	0.40	8.48	2.58	37.03	11.29	0.229	

Fig 5.5: Sensitivity Analysis Result of ELEPA – AKABEL Hydraulic Analysis.

8.6. OTHERS to SEIB to NUNR to ELEPA to AKABEL Pipeline Network System

There is need to increase the gas supply to 1 BCF. Sensitivity analysis was carried out on the pipeline system to confirm the suitable pipeline sizes that will deliver 1 BCF to Akabel at different pressure scenario as shown in result table 5.5 below.

OTHERS-SEIB-NUNR to ELEPA MANIFOLD to AKABEL Pipeline Hydraulic Analysis (1 BCF)																						
S/N	Simulation Case	Pipeline Size (in): OTHERS to SEIB	Pipeline Size (in): SEIB to NUNR	Pipeline Size (in): NUNR to ELEPA	Pipeline Size (in): ELEPA TO AKABEL (LOOP 1)	Pipeline Size (in): ELEPA TO AKABEL (LOOP 2)	Inlet Press (barg): OTHERS	Inlet Press (barg): SEIB	Inlet Press (barg): NUNR	Delivery Press (barg) at ELEPA MFD	Delivery Press (barg) at AKABEL (Loop 1_36")	Delivery Press (barg) at AKABEL (Loop 2_40")	Ertrotional Velocity Ratio (EVR<1, desired)					Velocity (ft/sec)				
													OTHERS to SEIB	SEIB to NUNR	NUNR to ELEPA	ELEPA TO AKABEL (LOOP 1)	ELEPA TO AKABEL (LOOP 2)	OTHERS to SEIB	SEIB to NUNR	NUNR to ELEPA	ELEPA TO AKABEL (LOOP 1)	ELEPA TO AKABEL (LOOP 2)
1	1 BCF Production	36	36	36	28	40	100	89.27	84.26	69.5	60.00	60.00	0.492	0.505	0.554	0.257	0.354	19.38	20.39	24.57	12.14	16.73
2							80	65.69	58.67	32.96	18.00	18.00	0.578	0.610	0.810	0.517	0.633	26.76	29.81	52.52	45.10	55.22
5		36	36	36	36	36	100	86.73	81.30	64.71	56.00	56.00	0.495	0.514	0.570	0.306	0.306	19.632	21.11	26.05	15	15.00
6							80	65.25	57.96	30.55	17.00	17.00	0.580	0.614	0.842	0.560	0.560	26.95	30.2	56.72	50.28	50.28
7		36	36	36	36	40	100	88.77	83.51	65.10	60.00	60.00	0.493	0.506	0.558	0.286	0.248	19.49	20.52	24.92	13.52	11.74
8							80	66.95	60.48	34.17	24.40	24.40	0.572	0.603	0.795	0.411	0.431	26.23	29.16	50.65	30.92	32.44
Pipeline Distance: OTHERS to SEIB is 40km																						

Pipeline Distance: OTHERS to SEIB is 40km

Fig 5.5: Sensitivity Analysis Result of OTHERS –SEIB – NUNR – ELEPA – AKABEL Hydraulic Analysis of 1 BCF.

The hydraulics considers an optimal arrival pressure that will require less power for compression at AKABEL with 2-loop production systems of already selected 36" pipeline and 40" from ELEPA to AKABEL as indicated in the table. The inlet pressures at the OTHER field are between 100barg to 80 barg.

9. Recommendations

The result of the hydraulics analysis of the Gas Gathering Pipeline Systems confirms and recommend the following pipeline sizes:

- BBOU to ELEPA 6.7km pipeline is recommended to be 24" since it satisfies the requirement of delivering the gas to ELEPA.
- 28" pipeline is recommended to deliver the peak production from OPUG to BBOU.
- For the gas transmission pipeline from ELEPA to AKABEL, 36" pipeline is recommended.

The recommended pipeline schematics is shown in fig 6.1.

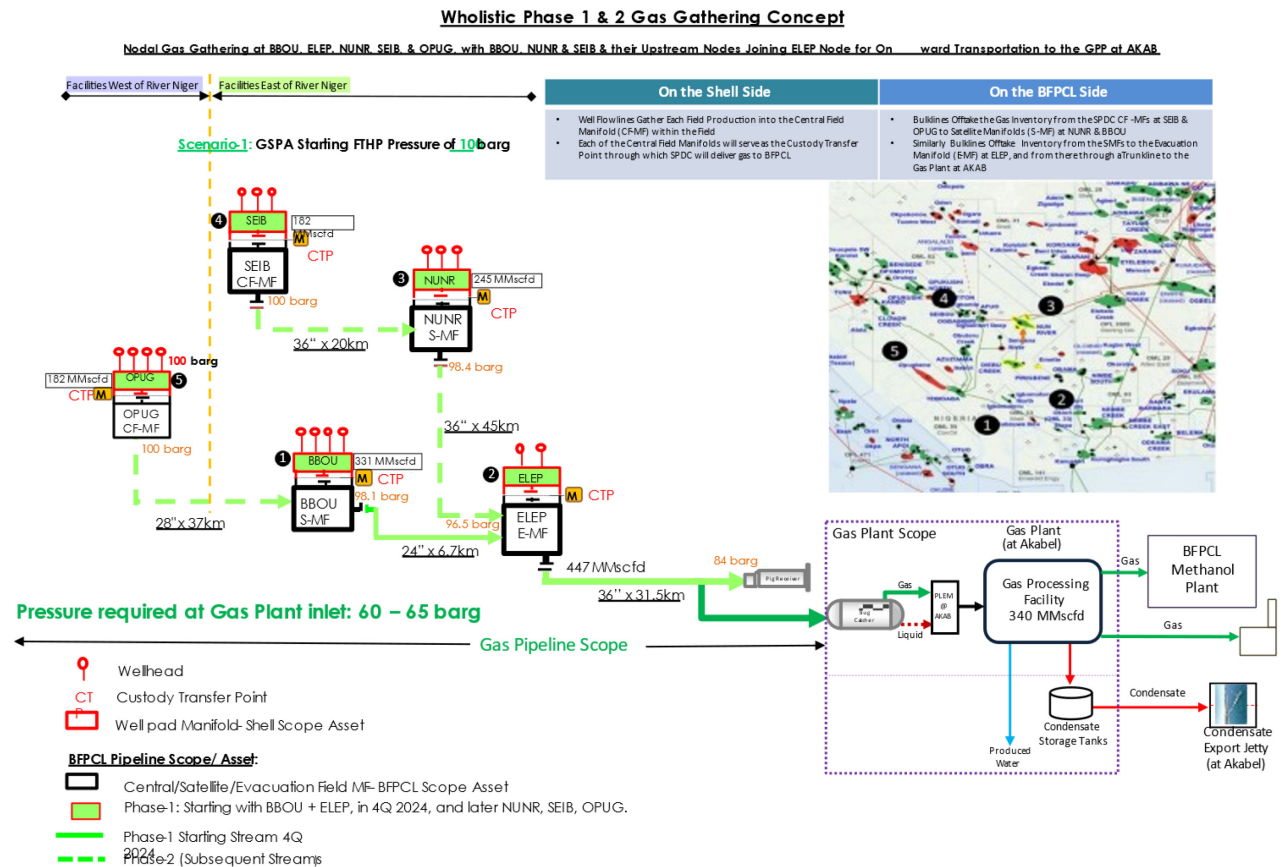


Fig 6.1: Recommended pipe